



Software Reliability Models

Lecture – 20



Software Reliability Models

- Software reliability models are mathematical models that aim to quantify the reliability of software systems based on historical data, metrics, or assumptions.
- These models provide insights into the expected reliability characteristics of the software and help in making informed decisions regarding testing, maintenance, and improvement strategies.



Failure Intensity Models

These models estimate the rate at which failures occur over time. Examples include the exponential distribution model and the nonhomogeneous Poisson process (NHPP) model.



Nonlinear Models

Nonlinear models capture the changing failure rates over time, considering factors such as the learning effect, debugging process, or environmental conditions.



Markov Models

Markov models represent the software system as a stochastic process with a set of states and transition probabilities.

These models assess the reliability based on the transitions between states.



Bayesian Models

Bayesian models incorporate prior knowledge and update reliability estimates based on observed failures and data.

These models are useful when limited data is available.



REFERENCE

"Software Quality Assurance: From Theory to Implementation" by Daniel Galin (2018)